

QIEMO QARQAN, China

WMO: 518550

Lat: 38.1329N

Lon: 85.5372E

Elev: 1248

StdP: 87.20

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-14.1	-24.5	0.5	-4.1	-22.6	0.6	-5.2	7.0	-2.5	5.9	-1.9	0.3	N/A	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	35.2	16.6	34.1	16.4	32.8	16.1	19.0	30.5	18.2	29.7	17.4	28.9	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	12.9	22.8	14.4	11.9	22.2	13.2	11.0	22.4	59.6	30.4	56.7	29.9	54.1	28.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.3	7.0	5.7		-18.6	38.5	2.2	1.2	-20.2	39.4	-21.5	40.1	-22.7	40.7	-24.4	41.6	
			DB	-19.3	21.7	1.9	1.7	-20.6	22.9	-21.8	23.9	-22.8	24.9	-24.2	26.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-6.7	-2.0	5.8	13.7	19.4	22.8	24.7	23.7	18.2	10.2	2.4	-4.9
	DBStd	11.49	3.01	3.14	4.20	3.72	3.28	2.65	2.16	2.40	2.71	3.74	4.07	3.48
	HDD10.0	1732	518	336	138	7	1	0	0	0	0	41	230	461
	HDD18.3	3397	777	569	389	147	26	2	0	1	34	252	479	719
	CDD10.0	1976	0	0	7	117	291	383	456	425	247	48	1	0
	CDD18.3	598	0	0	0	7	58	136	198	167	31	1	0	0
	CDH23.3	7925	0	0	0	189	847	1615	2415	2236	576	48	0	0
	CDH26.7	3496	0	0	0	59	297	692	1189	1097	153	9	0	0
Wind	WSAvg	1.7	1.3	1.7	2.2	2.4	2.2	1.8	1.8	1.9	1.8	1.6	1.2	1.0
Precipitation	PrecAvg	23	0	0	0	1	2	6	8	4	1	0	0	0
	PrecMax	48	5	3	8	19	19	17	30	16	5	2	5	2
	PrecMin	2	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	13	1	1	2	4	3	6	7	5	1	0	1	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	11.6	21.1	32.0	33.7	36.6	37.1	36.9	32.2	27.2	18.5	8.2
		MCWB	-1.3	2.6	7.3	13.2	14.2	18.0	18.1	17.2	15.1	11.3	6.9	0.9
	2%	DB	3.1	8.8	18.8	28.4	31.6	33.9	35.4	35.1	30.1	24.5	14.8	4.8
		MCWB	-2.5	0.8	6.2	11.5	14.1	16.7	17.1	16.3	14.1	10.1	5.1	-1.1
	5%	DB	1.2	6.9	16.6	25.2	29.9	32.3	34.1	33.9	28.5	21.9	12.1	2.9
		MCWB	-3.6	-0.2	5.5	10.1	13.8	16.0	16.9	16.0	13.2	9.2	3.9	-2.1
	10%	DB	-0.4	5.3	14.6	22.9	28.1	30.7	32.7	32.5	26.8	19.3	9.8	1.2
		MCWB	-4.6	-1.2	4.4	8.9	12.9	15.4	16.8	15.6	12.7	8.0	2.8	-3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	2.9	7.6	13.2	16.7	19.6	20.7	19.4	16.7	12.4	7.1	1.6
		MCDB	5.1	10.5	20.1	30.2	30.3	31.9	30.6	31.2	29.6	25.9	18.0	7.3
	2%	WB	-2.4	1.2	6.6	11.7	15.1	18.1	19.4	18.2	15.1	10.5	5.2	-0.6
		MCDB	2.8	8.3	17.9	27.1	29.4	30.7	30.5	30.6	27.9	23.2	13.8	4.1
	5%	WB	-3.5	-0.1	5.7	10.5	14.2	17.1	18.6	17.3	14.1	9.3	4.1	-1.8
		MCDB	1.1	6.5	16.3	24.4	27.9	29.5	29.7	30.0	25.9	21.1	11.6	2.8
	10%	WB	-4.5	-1.1	4.5	9.3	13.3	16.3	17.9	16.6	13.2	8.3	2.9	-2.9
		MCDB	-0.5	5.0	14.0	21.8	26.4	28.1	28.7	29.3	24.7	18.7	9.6	1.2
Mean Daily Temperature Range		MDBR	12.0	12.9	13.8	14.8	14.1	13.5	13.6	15.2	16.2	15.8	13.2	11.2
	5% DB	MCDBR	13.9	15.2	16.7	18.8	17.9	16.9	17.3	18.5	18.7	18.7	16.4	13.5
		MCWBR	9.9	9.3	8.4	7.9	7.2	6.2	5.3	6.1	7.0	8.5	9.2	9.0
		MCDBR	13.9	14.5	15.8	17.8	15.3	13.5	12.9	14.9	16.4	18.2	16.0	13.2
	5% WB	MCWBR	9.8	9.0	8.0	7.6	6.4	5.9	5.1	5.7	6.9	8.4	9.0	8.8
Clear-Sky Solar Irradiance	taub	0.343	0.407	0.505	0.600	0.628	0.646	0.628	0.613	0.568	0.455	0.378	0.350	
	taud	2.060	1.898	1.677	1.511	1.513	1.529	1.592	1.602	1.649	1.817	1.984	2.039	
	Ebn at Noon	803	788	743	702	694	681	689	684	681	724	747	757	
	Edh at Noon	120	163	225	282	286	282	263	254	229	174	129	115	
All-Sky Solar Radiation	RadAvg	2.81	3.76	5.17	6.18	6.85	6.91	6.74	6.24	5.65	4.64	3.29	2.58	
	RadStd	0.30	0.20	0.24	0.31	0.32	0.33	0.37	0.40	0.19	0.11	0.15	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	-0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-139	-117

Nomenclature: See separate page